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Graded Assignment • 2h

Due Apr 1, 11:59 PM IST

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1. If you are querying data with Amazon Athena, you can use AWS Lake Formation to simplify how you secure and connect to your data from Amazon QuickSight. After Lake Formation is configured, you can use Amazon QuickSight to access databases and tables. 1 / 1 point

Which method can you use to access databases or tables through SQL queries? (Select TWO.)

- [ ] Use the AWS Glue service used for building the data lake.
- [x] Use the Amazon Athena Console, the AWS CLI, or your favorite query editor.

Correct
If you are querying data with Amazon Athena, you can use AWS Lake Formation to simplify how you secure and connect to your data from Amazon QuickSight. You can use a variety of methods to access databases or tables through SQL queries. You can use the Amazon Athena Console, the AWS CLI, or your favorite query editor. You can also use the full-featured editor provided by Amazon QuickSight where you can write SQL queries.

For more information, refer to the "Querying Data with Amazon Athena" lesson in Module 2: Data Ingestion, Cataloging, and Preparation.

- [x] Use the full-featured editor provided by Amazon QuickSight where you can write SQL queries.

Correct
If you are querying data with Amazon Athena, you can use AWS Lake Formation to simplify how you secure and connect to your data from Amazon QuickSight. You can use a variety of methods to access databases or tables through SQL queries. You can use the Amazon Athena Console, the AWS CLI, or your favorite query editor. You can also use the full-featured editor provided by Amazon QuickSight where you can write SQL queries.

For more information, refer to the "Querying Data with Amazon Athena" lesson in Module 2: Data Ingestion, Cataloging, and Preparation.

- [ ] Use the reports generated by Amazon Athena and parse it in a relational DB for running SQL queries.
- [ ] Use the AWS Lambda function to execute SQL queries on the data in the Lake Formation data lake.

2. Which AWS Glue feature can determine the schema of your data? 1 point

- ( ) Glue Job
- ( ) Classifier
- ( ) Development Endpoint
- (•) Crawler

Incorrect
AWS Glue Crawlers are designed to discover and catalog data by inferring schemas, while other features like Glue Jobs, Classifiers, and Development Endpoints serve different purposes such as data processing, classifying data, and providing interactive development environments.

3. You need to author an ETL job in AWS Glue. Your datasets are in Amazon S3. Your goal is to filter, join, and aggregate two different datasets. What tool can you use to create and run AWS Glue job scripts that could help you achieve this goal? (Select TWO.) 0.8 / 1 point

- [x] AWS Glue Script Editor

Correct
AWS Glue runs Python shell scripts or PySpark or Scala scripts. You can create your scripts manually or use AWS Glue resources like the AWS Glue Studio and AWS Glue Script Editor to support your script development, visualization, and testing.

For more information, refer to the "Data Processing with AWS Glue" lesson in Module 4: Data Analytics and Processing.

- [ ] AWS Glue Crawler
- [ ] AWS Glue Triggers
- [x] AWS Glue Studio

This should not be selected
AWS Glue runs Python shell scripts or PySpark or Scala scripts. You can create your scripts manually or use AWS Glue resources like the AWS Glue Studio and AWS Glue Script Editor to support your script development, visualization, and testing.

For more information, refer to the "Data Processing with AWS Glue" lesson in Module 4: Data Analytics and Processing.

- [ ] AWS Glue Data Catalog

4. Which combination of AWS services form the storage layer for a data lake in AWS? 1 / 1 point

- ( ) Amazon S3, AWS Glue, Amazon DynamoDB
- (•) Amazon S3, Amazon RDS, Amazon Redshift
- ( ) Amazon RDS, Amazon DynamoDB, Amazon EMR
- ( ) Amazon EC2, Amazon RDS, Amazon DynamoDB

Correct
You can store all your relational and non-relational data in the data lake. Amazon S3, Amazon RDS, Amazon Redshift form the storage layer for a data lake in AWS. Data lakes can store exabytes of data and scale with your needs for storage. You can do storage and deep analytics on your data using data lakes.

You can find more information in the "Components and architectures" lesson in Module 1: Introduction to Data Lakes.

5. Shirley is a data administrator who needs to set up a data lake for their company, AnyCompany. Currently, all data is stored in Amazon S3. John is a marketing manager and needs write access to customer purchasing information. Shirley registers the Amazon S3 path containing customer purchasing information with AWS Lake Formation and grants John access to the Amazon S3 path. John tries to create a database and gets an error. What could be the possible issue here? 1 / 1 point

- ( ) A bucket policy is preventing John from creating a database
- ( ) John is missing data lake administrator permissions.
- (•) Shirley also needs to grant John permission to create databases
- ( ) Shirley doesn't have sufficient permission for accessing Amazon S3

Correct
For any granular-level action on a database, like CREATE_TABLE, John needs permissions to call this action on the database. Therefore, Shirley also need to grant John permission to create databases in the data lake.

For more information, refer to the "Security and access controls" lesson in Module 5: AWS Lake Formation Additional Configurations.

6. AWS Lake Formation provides machine learning capabilities to create custom transforms to cleanse your data. There is currently one available transform named FindMatches. You can create these transforms when you create a job. After the transform, where is the data stored? 1 point

- ( ) A user defined data source
- (•) User selected location in S3
- ( ) Amazon SageMaker
- ( ) AWS Glue

Incorrect
While the transformed data can be stored in an Amazon S3 location that you select when creating the AWS Glue job in Lake Formation, this is not the correct answer for this scenario.

7. While creating a blueprint in AWS Lake Formation, John received the following error: 1 / 1 point
"User: is not authorized to perform: iam:PassRole on resource"

How can this be fixed? (Select TWO.)

- [ ] John's account is disabled. Enable the user account.
- [ ] Grant John Read Only permission on the S3 data source
- [x] Ask John to choose a different role with the required passrole permissions

Correct
Permissions in Lake Formation are based on roles or personas. In order to create a blueprint in AWS Lake Formation, John would need the passrole permissions. This can be done either by updating his IAM policy with pass role permissions or by asking John to choose a different role with the required passrole permissions.

For more information, see the "Security and access controls" lesson in Module 5: AWS Lake Formation Additional Configurations.

- [x] Update John's IAM policy to be able to pass the role

Correct
Permissions in Lake Formation are based on roles or personas. In order to create a blueprint in AWS Lake Formation, John would need the passrole permissions. This can be done either by updating his IAM policy with pass role permissions or by asking John to choose a different role with the required passrole permissions.

For more information, see the "Security and access controls" lesson in Module 5: AWS Lake Formation Additional Configurations.

- [ ] Ask John to choose a different AWS Region

8. A customer is planning to schedule and implement a complex multi job extract, transform and load (ETL) activity. They want to track the activity as a single entity. How should the customer proceed? 1 / 1 point

- ( ) Use an Administrator IAM User and use that account for scheduling the ETL Job
- ( ) Use a database in the data lake's Data Catalog and store the ETL Job information in it
- (•) Use a workflow that defines the data source and schedule to import data into your data lake
- ( ) Use manually created Glue crawlers and run it on demand or on a schedule

Correct
An easy way for the customer to track a complex multi job extract, transform and load (ETL) activity as a single entity is to use a workflow that defines the data source and schedule to import data into their data lake. A workflow is a container for a set of related AWS Glue jobs, crawlers, and triggers. Lake Formation can track the status of a workflow as a single entity.

For more information, refer to the "Using Blueprints and Workflows" lesson in Module 5: AWS Lake Formation Additional Configurations.

9. In which scenario would a data scientist use AWS Glue jobs? (Select TWO.) 0.8 / 1 point

- [ ] Assigning data in real time as data comes into the data lake
- [ ] Transforming data in near real time as data comes into the data lake
- [x] Assigning data in batches on schedule or on demand

This should not be selected
While Glue can be used to prepare data for analysis, it is not the best tool for ad-hoc, interactive data explorations. Tools like Amazon QuickSight or Jupyter Notebooks would be better suited for this.

- [x] Transforming data in batches on schedule or on demand

Correct
AWS Glue Streaming ETL can process near real-time data. AWS Glue ETL is typically run in batches.

- [ ] Developing machine learning models

10. What is the AWS Glue Data Catalog? 1 / 1 point

- ( ) A fully managed extract, transform, and load (ETL) pipeline service
- ( ) A service to schedule jobs
- ( ) A visual data preparation tool
- (•) An index to the location, schema, and runtime metrics of your data

Correct
The AWS Glue Data Catalog contains references to data used as sources and targets of your ETL jobs in AWS Glue. To create your data warehouse or data lake, you must catalog this data. The AWS Glue Data Catalog is an index to the location, schema, and runtime metrics of your data. You use Data Catalog information to create and monitor your ETL jobs. Information in the Data Catalog is stored as metadata tables, where each table specifies a single data store. Typically, you run a Crawler to take inventory of the data in your data stores, but there are other ways to add metadata tables into your Data Catalog. The runtime metrics are the business operations information regarding the resources in the catalog.